6. Appendix

A. ESP32 CODE

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <WiFiClientSecure.h>

const char* ssid     = "test"; const char* password = "123456789"; const char* apiUrl
= "https://adstnqux35or2vfzoz2zpquoka0vwfkc.lambda-url.us-east-
1.on.aws/";

#define RXD2 16
#define TXD2 17

bool flameAlertSent      = false; bool
suppressedAlertSent = false;

void sendAlert(String type) {
WiFiClientSecure client;
client.setInsecure();   HTTPClient
http;   http.begin(client, apiUrl);
  http.addHeader("Content-Type", "application/json");
String body = "{\"type\":\"" + type + "\"}";   int code =
http.POST(body);
  Serial.println("HTTP: " + String(code) + " type: " + type);   http.end();
}
```

```
void setup() {
  Serial.begin(115200);
  Serial2.begin(9600, SERIAL_8N1, RXD2, TXD2);
  WiFi.begin(ssid, password);
  Serial.print("Connecting WiFi");   while
(WiFi.status() != WL_CONNECTED) {
delay(500);
    Serial.print(".");
  }
  Serial.println("\nWiFi Connected!");
}

void    loop()    {          if
(Serial2.available()) {
   String       msg       =       Serial2.readStringUntil('\n');
msg.trim();
   Serial.println("From Arduino: " + msg);

   if  ((msg  ==  "SPRAY_RIGHT"  ||                  msg  ==
"SPRAY_LEFT"   ||             msg  ==  "SPRAY_FRONT"  ||
msg == "MOVE_RIGHT"  ||          msg == "MOVE_LEFT"
||         msg == "MOVE_FORWARD") && !flameAlertSent)
{        sendAlert("flame");        flameAlertSent       = true;
suppressedAlertSent = false;

   }
```

```
    if (msg == "NO_FIRE" && flameAlertSent && !suppressedAlertSent) {
sendAlert("suppressed");       suppressedAlertSent = true;
flameAlertSent      = false;
    }
  }
}
```

B. Arduino CODE

```
#include <Servo.h>
```

```
// ================= MOTOR DRIVER =================
const int ENA = 5; const int IN1 = 13; const int IN2 = 11; const int
IN3 = 10; const int IN4 = 9; const int ENB = 6;
// ================= FLAME SENSORS =================
const int RIGHT_SENSOR = A0; const int FRONT_SENSOR = A1;
const int LEFT_SENSOR  = A3;
// ================= OUTPUTS =================
const int BUZZER = 8; const int PUMP   = 7;
// ================= SERVO =================
Servo myServo; const int SERVO_PIN = 4;
// ================= MOTOR SPEED ================= const
int motorSpeed = 150;

// baseline no fire readings
const int R_BASE = 1000;
const int F_BASE = 480; const
int L_BASE = 1000;

// detect far fire — small drop = fire
const int fireThresh = 50;
// stop at safe distance — bigger drop = near const
int nearThresh = 150;

void setup() {   pinMode(ENA,
OUTPUT);   pinMode(IN1,
OUTPUT);   pinMode(IN2,
OUTPUT);   pinMode(IN3,
OUTPUT);   pinMode(IN4,
```

```
OUTPUT);   pinMode(ENB,
OUTPUT);   pinMode(BUZZER,
OUTPUT);   pinMode(PUMP,
OUTPUT);
digitalWrite(BUZZER, LOW);
digitalWrite(PUMP, LOW);
myServo.attach(SERVO_PIN);
myServo.write(90);
Serial.begin(9600);   stopCar();
}

void loop() {
  int R = analogRead(RIGHT_SENSOR);
int F = analogRead(FRONT_SENSOR);
int L = analogRead(LEFT_SENSOR);

  int Rdrop = R_BASE - R;
int Fdrop = F_BASE - F;   int
Ldrop = L_BASE - L;


  bool Rfire = Rdrop > fireThresh;
bool Ffire = Fdrop > fireThresh;   bool
Lfire = Ldrop > fireThresh;

  bool Rnear = Rdrop > nearThresh;
bool Fnear = Fdrop > nearThresh;   bool
Lnear = Ldrop > nearThresh;
```

```
  Serial.print("R:"); Serial.print(R);
  Serial.print(" F:"); Serial.print(F);
  Serial.print(" L:"); Serial.print(L);
  Serial.print(" | Rd:"); Serial.print(Rdrop);
  Serial.print(" Fd:"); Serial.print(Fdrop);
  Serial.print(" Ld:"); Serial.println(Ldrop);

  // ================================================
  // NO FIRE
  // ================================================
if (!Rfire && !Ffire && !Lfire) {     stopCar();     digitalWrite(PUMP,
LOW);     digitalWrite(BUZZER, LOW);     myServo.write(90);
    Serial.println("NO_FIRE");
  }

  // ================================================
  // RIGHT FIRE
  // ================================================
else if (Rfire && Rdrop >= Fdrop && Rdrop >= Ldrop) {
digitalWrite(BUZZER, HIGH);     if (Rnear) {
      // near — stop immediately and spray
stopCar();       myServo.write(45);
digitalWrite(PUMP, HIGH);
      Serial.println("SPRAY_RIGHT");
    } else {
      // far — move toward fire
myServo.write(45);       // aim while moving
digitalWrite(PUMP, LOW);       turnRight();
```

```
    Serial.println("MOVE_RIGHT");
   }
 }

 // ================================================
 // LEFT FIRE
 // ================================================
else if (Lfire && Ldrop >= Fdrop && Ldrop >= Rdrop) {
digitalWrite(BUZZER, HIGH);
   if (Lnear) {
stopCar();
myServo.write(135);
digitalWrite(PUMP,
HIGH);
    Serial.println("SPRAY_LEFT");
   } else {      myServo.write(135);      // aim
while moving      digitalWrite(PUMP, LOW);
turnLeft();
    Serial.println("MOVE_LEFT");
   }
 }

 // ================================================
 // FRONT FIRE
 //       ================================================
else if (Ffire) {
   digitalWrite(BUZZER, HIGH);
if (Fnear) {      stopCar();
```

```
myServo.write(90);
digitalWrite(PUMP, HIGH);
      Serial.println("SPRAY_FRONT");
   } else {       myServo.write(90);       // aim
while moving       digitalWrite(PUMP, LOW);
moveForward();
      Serial.println("MOVE_FORWARD");
    }
  }

  else {     stopCar();
digitalWrite(PUMP, LOW);
digitalWrite(BUZZER, LOW);
myServo.write(90);
    Serial.println("NO_FIRE");
  }

  delay(500);
}

// ================= FORWARD =================
void moveForward() {   analogWrite(ENA, motorSpeed);
analogWrite(ENB, motorSpeed);   digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);   digitalWrite(IN3, LOW);
digitalWrite(IN4, HIGH);
}
// ================= RIGHT =================
```

```
void turnRight() {
analogWrite(ENA, motorSpeed);
analogWrite(ENB, motorSpeed);
digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);
digitalWrite(IN3, HIGH);
digitalWrite(IN4, LOW);
}
// ================= LEFT =================
void turnLeft() {
analogWrite(ENA, motorSpeed);
analogWrite(ENB, motorSpeed);
digitalWrite(IN1, LOW);
digitalWrite(IN2, HIGH);
digitalWrite(IN3, LOW);
digitalWrite(IN4, HIGH);
}
// ================= STOP =================
void stopCar() {
analogWrite(ENA, 0);
analogWrite(ENB, 0);
digitalWrite(IN1, LOW);
digitalWrite(IN2, LOW);
digitalWrite(IN3, LOW);
digitalWrite(IN4, LOW);
}
```